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Android Ransomware-like Activity Early Warning in a Controlled Lab

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Problem Description

- **Signature-based antivirus tools** may fail to detect zero-day ransomware whose signatures have not yet been catalogued.
- **Delayed detection** allows ransomware to modify, rewrite, or rename multiple files before an alert is triggered.
- **Single-feature detection approaches** may miss suspicious behavior when individual indicators appear benign in isolation.
- **Diverse behavioral indicators** such as file-access patterns, entropy changes, and API activity require specialized analysis.
- **Need for early warning**: A unified detection mechanism is required to combine multiple behavioral signals and identify potentially malicious activity before extensive file damage occurs.
- **Challenge**: Develop a four-layer ensemble system that processes a common telemetry dataset, generates individual maliciousness scores, and combines them into a final risk score for timely detection.

That is the threat RANSCOPE is built to stop.

Background

- **Growing ransomware threat**: Ransomware can rapidly modify, rewrite, or rename files, potentially causing significant data loss before users become aware of the attack.
- **Limitations of signature-based detection**: Traditional antivirus tools may fail to identify previously unknown ransomware whose signatures are not yet available.
- **Behavioral indicators**: Ransomware-like activity can be identified through suspicious patterns, such as rapid file access, repeated write and rename operations, changes in file entropy, and unusual API activity.
- **Android monitoring constraints**: Android sandboxing and application permissions can limit access to system-level information, making controlled and well-defined monitoring environments important.
- **Need for early warning**: Detecting suspicious behavior before extensive file modifications occur can reduce potential damage.
- **Proposed optimization**: RANSCOPE uses a four-layer ensemble approach-MPVG, semantic file access, entropy acceleration, and cognitive API intent to analyze different behavioral indicators and combine their individual risk scores into a unified risk score for early warning.

Literature & RFC Review

- Android Ransomware Research [1–4]: Existing studies explore Android ransomware detection through behavioural analysis, system API usage, system-call monitoring and early-stage threat detection. These approaches motivate combining multiple behavioural indicators for earlier warnings.
- Mobile Security Standards [5–6]: OWASP MASVS defines mobile application security requirements, while MASTG provides practical security testing guidance. They inform the secure design and evaluation of the RANSCOPE-M prototype.
- RFC 9424 – Indicators of Compromise [7–8]: Explains how IoCs help identify and defend against cyberattacks. RANSCOPE-M extends this concept by combining simulated behavioural signals rather than relying solely on known indicators.
- Research Gap & Proposed Innovation: RANSCOPE-M proposes four specialized layers—MPVG (permission patterns), SFIM (file-access behaviour), EAP (entropy changes) and CAIT (API-call sequences). A weighted ensemble combines their outputs into a consolidated risk score for early warning.
- Testing & Evaluation: The prototype uses Android Studio-generated simulated telemetry, two benign workloads and three harmless ransomware-like simulations. Performance will be assessed using detection delay, files changed before alert, precision and recall.

Proposed Solution

- **Android Studio Test Environment**: Develop RANSCOPE in an Android Studio emulator using synthetically generated behavioural telemetry and simulated ransomware-like activities within a controlled test directory.
- **Simulated Behavioural Monitoring**: Generate and capture file-system event logs, including file opens, writes, renames and deletes, to identify suspicious file-rewriting patterns without using real ransomware.
- **Parallel Multi-Layer Detection**: Process the generated telemetry through four specialized pretrained models: MPVG, SFIM, EAP, CAIT
- **Layer-Specific Feature Extraction**: Each model selectively processes the features relevant to its detection task from the generated dataset, enabling all four models to analyse the same test scenario in parallel.
- **Weighted Ensemble Risk Scoring**: Each model generates a risk score between 0 and 1. The four scores are combined using predefined layer weights to calculate a unified ransomware-risk score.
- **Early-Warning and Alert Generation**: Apply a calibrated threshold to the ensemble score to flag suspicious activity, recording the alert timestamp and the number of files modified before detection.
- **Performance Evaluation**: Test the system using two benign workloads and three harmless ransomware-like simulations. Evaluate detection delay, files changed before alert, precision and recall using the generated telemetry logs and ground-truth labels.

Optimization Proposed

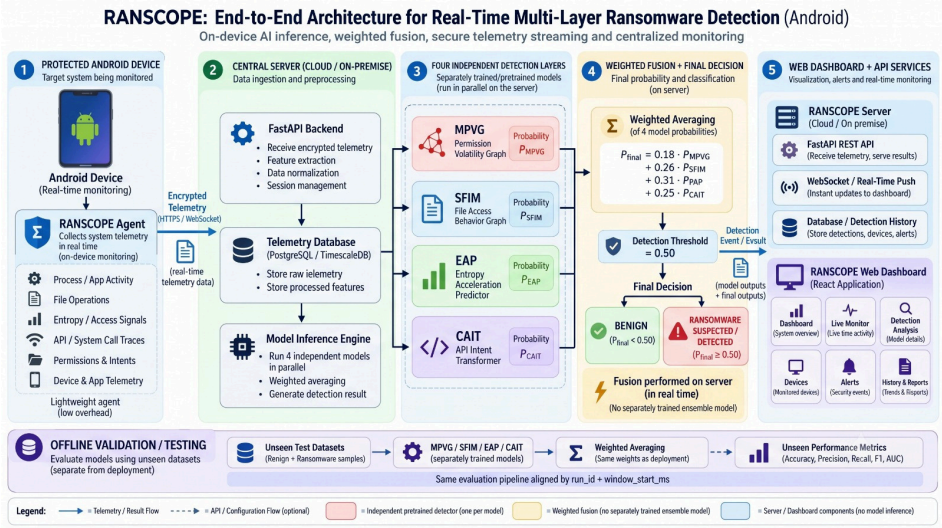
Aspect	Before Optimization	After Optimization
Detection Approach	Reliance on individual behavioural indicators.	Four specialized models analyse complementary behavioural signals in parallel.
Feature Processing	Limited separation of features by detection task.	Layer-specific feature extraction processes only the relevant features for each model.
Risk Assessment	Individual models produce independent predictions.	Weighted ensemble combines all four risk scores into one unified risk score.
Detection Timing	Suspicious activity may be detected after substantial file modification.	Early-warning logic aims to flag suspicious activity before extensive file damage.
Monitoring	Limited visibility into file-operation sequences.	Structured telemetry logs capture simulated file opens, writes, renames and deletes.

Optimization Proposed

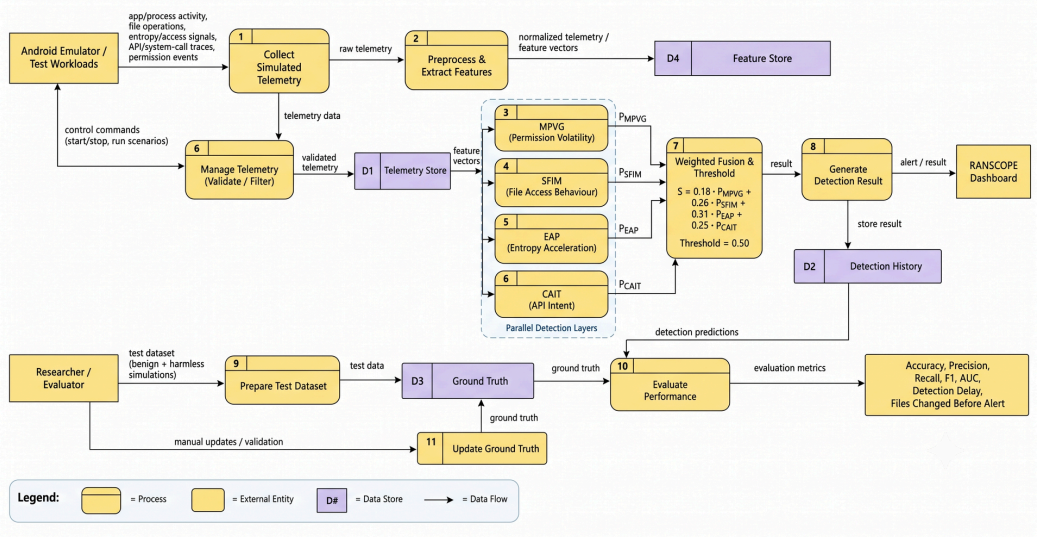
Aspect	Before Optimization	After Optimization
Threat Response	Detection without a clearly defined controlled response.	Alerts trigger termination of only the designated simulated test workload.
Testing Environment	Limited reproducibility of behavioural scenarios.	Android Studio enables repeatable benign and ransomware-like simulations
Performance Evaluation	Detection effectiveness may be assessed using limited metrics.	Evaluation measures detection delay, files changed before alert, precision and recall.

Key enhancement: RANSCOPE combines parallel multi-layer detection, weighted risk fusion and controlled early intervention to improve the systematic detection and evaluation of ransomware-like activity

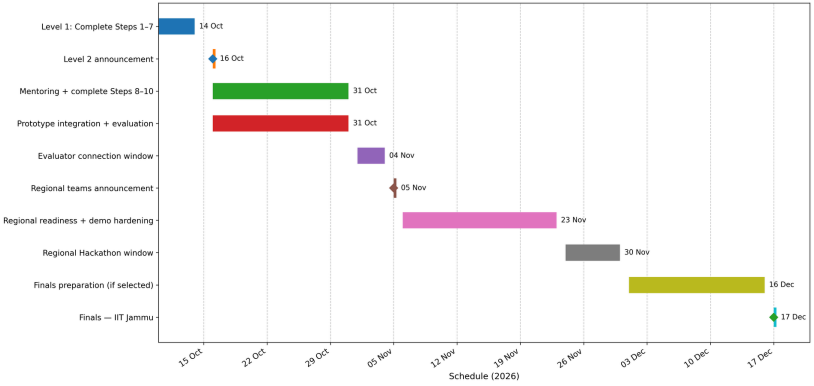
Solution Architecture & Design



Data Flow Diagram - Ranscope



Deployment Timeline



Work methodology

Step 1 — Develop

Build the Android Studio prototype and generate repeatable telemetry for two benign workloads and three harmless ransomware-like simulations.

Step 2 — Integrate

Implement file-system monitoring and, where available, feed the relevant telemetry into the four specialized models and weighted ensemble.

Step 3 — Validate

Measure detection delay, files changed before alert, precision and recall against known simulation labels. Record results in reproducible logs.

Step 4 — Demonstrate

Present the controlled test environment, simulated attack detection, alert generation, safe workload termination and measured results.

Bibliography

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Thank You !